Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A large tank filled with water to a height ' h ' is to be emptied through a small hole at the bottom. The ratio of time taken for the level of water to fall from h to $\frac{h}{2}$ and $\frac{h}{2}$ to zero is

(A) $\sqrt{2}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}-1$

(D) $\frac{1}{\sqrt{2}-1}$

Q.2

A large container (with open top) of neglect mass and uniform cross-sectional area. A has a small hole of cross-sectional area a in its side wall near the bottom. The container is keep on a smooth horizontal platform and contains a liquid of density ρ and mass m. If the
liquid starts flowing out of the hole at time $t=0$, the initial acceleration of the container is

(A) $\frac{\mathrm{ga}}{\mathrm{A}}$

(B) $\frac{gA}{a}$

(C) $\frac{2g\mathrm{a}}{\mathrm{A}}$

(D) $\frac{gA}{2a}$

Q.3

A wide vessel of uniform cross-section with a small hole in the bottom is filled win
$40\ \mathrm{cm}$ thick layer of water and $30\ \mathrm{cm}$ thick layer of kerosene. The relative density of
kerosene is 0.8. The initial velocity of flow of water streaming out of the hole is (take $g=10\ \mathrm{ms}^{-2}$)

(A) $\frac{2}{\sqrt{5}}\ \mathrm{ms}^{-1}$

(B) $\frac{4}{\sqrt{5}}\ \mathrm{ms}^{-1}$

(C) $\frac{6}{\sqrt{5}}\ \mathrm{ms}^{-1}$

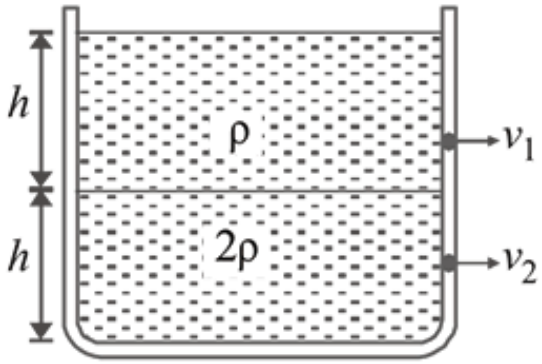
(D) $\frac{8}{\sqrt{5}}\ \mathrm{ms}^{-1}$

Q.4

Equal volumes of two immiscible liquids of densities ρ and 2ρ are filled in a vessel as shown in figure. Two

@IITJEE_Advanced

small holes are punched at depth $h/2$ and $3h/2$ from the surface of lighter liquid. If v_1 and v_2 are the velocities of a flux at these two holes, then v_1/v_2 is



(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{\sqrt{2}}$

Q.5

A long cylindrical vessel is half-filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of the vessel is $5\ \mathrm{cm}$ and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be

(A) 0.4

(B) 2.0

(C) 0.1

(D) 1.2

Q.6

A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density ρ_1 $(\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, i.e., $\mathrm{F}_{\text{viscous}} = -kv^2(k > 0)$. The terminal speed of the ball is

(A) $\sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

(B) $\frac{Vg\rho_1}{k}$

(C) $\sqrt{\frac{Vg\rho_1}{k}}$

(D) $\frac{Vg(\rho_1-\rho_2)}{k}$

Q.7

@IITJEE_Advanced

Two capillary of length Land 2L and of radius R and $2\mathrm{R}$ are connected in series. The net rate of flow of fluid through them will be (given rate to the flow through single capillary, $\mathrm{X} = \frac{\pi \mathrm{PR}^4}{8\eta \mathrm{L}}$)

(A) $\frac{8}{9}X$

(B) $\frac{9}{8}X$

(C) $\frac{5}{7}X$

(D) $\frac{7}{5}X$

Q.8

A small spherical ball of radius r falls freely under gravity through a distance h before entering a tank of water. If, after entering the water, the velocity of the ball does not change, then h is proportional to

(A) r^2

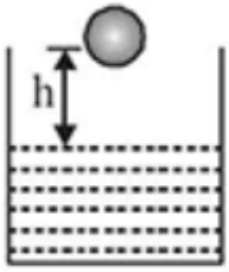
(B) r^3

(C) r^4

(D) r^5

Q.9

A ball of radius r and density ρ falls freely under gravity through a distance h before entering water. Velocity of ball does not change even on entering water. If viscosity of water is η the value of h is given by



(A) $\frac{2}{9}r^2\left(\frac{1-\rho}{\eta}\right)g$

(B) $\frac{2}{81}r^2\left(\frac{\rho-1}{\eta}\right)g$

(C) $\frac{2}{81}r^4\left(\frac{\rho-1}{\eta}\right)^2 g$

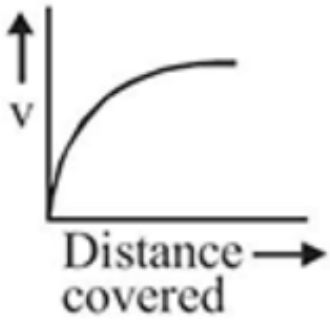
(D) $\frac{2}{9}r^4\left(\frac{\rho-1}{\eta}\right)^2 g$

Q.10

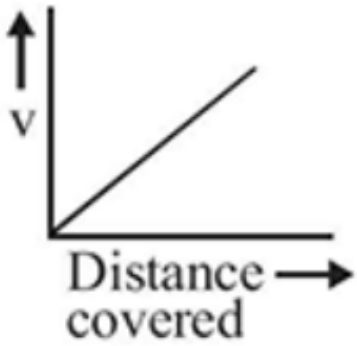
A lead shot of $1\ \mathrm{mm}$ diameter falls through a long column of glycerine. The variation of its velocity v with distance covered is represented by

@IITJEE_Advanced

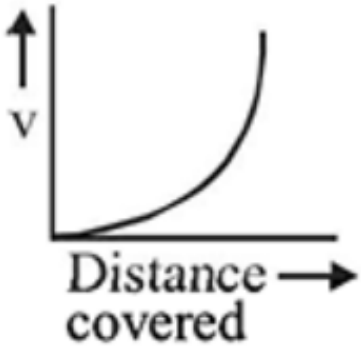
(A)



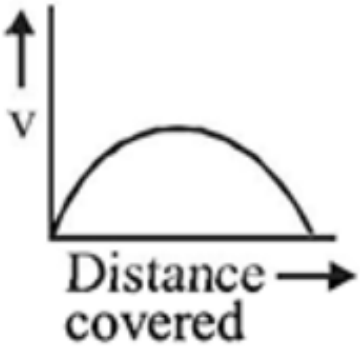
(B)



(C)



(D)



Q.11

Two liquids of densities d_1 and d_2 are flowing in identical capillary tubes uder the same pressure difference. If t_1 and t_2 are time taken for the flow of equal quantities (mass) of liquids, then the ratio of coefficient of viscosity of liquids must be

(A) $\frac{d_1 t_1}{d_2 t_2}$

(B) $\frac{t_1}{t_2}$

(C) $\frac{d_2}{d_1}\frac{t_2}{t_1}$

@IITJEE_Advanced

(D) $\sqrt{\frac{d_1 t_1}{d_2 t_2}}$

Q.12

If the terminal speed of a sphere of gold (density $= 19.5\ \mathrm{kg/m^3}$) is $0.2\ \mathrm{m/s}$ in a viscous liquid (density $= 1.5\ \mathrm{kg/m^3}$), find the terminal speed (in $\mathrm{m/sec}$) of a sphere of silver (density = $10.5\ \mathrm{kg/m^3}$) of the same size in the same liquid.

(A) 0.2

(B) 0.3

(C) 0.4

(D) 0.1

Q.13

A rectangular tank is filled to the brim with water. When a hole at its bottom is unplugged, the tank is emptied in time T. If the tank is half-filled with water, it will be emptied in time

(A) $\frac{T}{\sqrt{2}}$

(B) $\frac{T}{\sqrt{3}}$

(C) $\frac{T}{2}$

(D) $\frac{T}{2\sqrt{2}}$

Q.14

A wooden ball of density σ is released from the bottom of a tank which is filled with a liquid of density ρ; $(\rho > \sigma)$ up to a height h_1. The ball rises in the liquid, emerges from its surface and attains a height h_2 in air. If viscous effects are neglected, the ratio h_2/h_1 is

(A) $\left(\frac{\rho}{\sigma} + 1\right)$

(B) $\left(\frac{\rho}{\sigma} - 1\right)$

(C) $\frac{\rho}{\sigma}$

(D) $\frac{\sigma}{\rho}$

Q.15

A cylindrical vessel of a large crosssectional area has two tiny holes A and B and lies on a horizontal floor. Hole A is at a depht h_1 below the free surface of liquid in the vessel and hole B is at a height h_2 above the bottom of the vessel. When the holes are unplugged, the streams emerging from A and B strike the floor at the same point. The ratio h_1/h_2 is

(A) $2\sqrt{2}$

@IITJEE_Advanced

(B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) 1

Answers & Solutions

Q.1 Answer:

$\sqrt{2}-1$

Solution:

$t=k\left(\sqrt{h_1}-\sqrt{h_2}\right)$
$\therefore t_1=k(\sqrt{h}-\sqrt{h/2})$ **and** $t_2=k(\sqrt{h/2}-0)$
$\therefore \frac{t_1}{t_2}=\sqrt{2}-1$

Q.2 Answer:

$\frac{2ga}{A}$

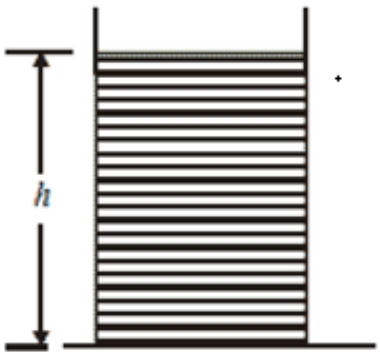
Solution:

Let h be the initial height of the liquid of dersity p in the container of cross-sectional area A. The mass of the liquid in the container initially is

$\mathrm{m}=$ **Ahp**

From Torricelli's theorem, the velocity of the liquid flowing out of the hole is $v\sqrt{2\ ?}$

$\therefore$ Volume of liquid flowing out per unit time $=\mathrm{av}.$



Hence the mass of liquid flowing out per unit time = pav. Therefore, the momentum carried per unit time by the liquid Flowing out is = (mass per unit time) x velocity $=(\rho av)v=pav^2$

This is the rate of change of momentum of the liquid flowing out which is the force with which the liquid flows out at $\mathrm{t}=0.$

$\therefore$ Initial acceleration $=\frac{\text{force}}{\text{mass}}$

=\frac{\text { pav }^{2}}{\text { Ah\rho }}=\frac{q u^{2}}{\mathrm{Ah}}

$=\frac{\mathrm{a}\times 2\mathrm{gh}}{\mathrm{Ah}}\left(\because \mathrm{v}=\sqrt{2\mathrm{gh}}\right)$

$=\frac{2\mathrm{ga}}{\mathrm{A}}$

Q.3 Answer:

@IITJEE_Advanced

$\frac{8}{\sqrt{5}}\ \mathrm{ms}^{-1}$

Solution:

Let ρ_w and ρ_k be the densities of water and kerosene. The initial weight of the liquid in the vessel $\mathrm{h_w}\rho_\mathrm{w}\mathrm{ag} + \mathrm{h_k}$ pr ag where $\mathrm{h_w}$ and $\mathrm{h_k}$ are the thicknesses of water and kerosene layers and a is the cross-sectional area of the vessel. Let this weight be equivalent to water layer of thickness h, then

$$hp_w ag = h_w \rho_w ag + h_k p_k ag$$

or $h = \left\{ h_w + h_x \left(\frac{p_1}{A_n} \right) \right\}$

$= 0.4 + 0.3 \times 0.8 = 0.64\ \mathrm{m}$

From Torricelli's theorem, the velocity of effiux is

$\nabla = \sqrt{2\,\mathrm{th}}$

$\nabla = \sqrt{2 \times 10 \times 0.04} = \frac{8}{\sqrt{5}}\ \mathrm{ms}^{-1}$

Q.4 Answer:

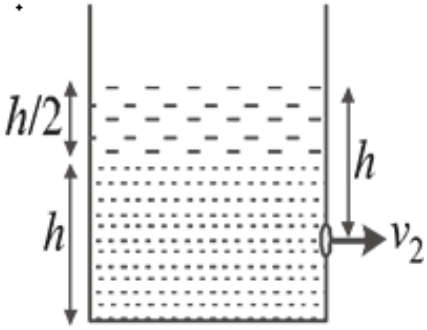
$\frac{1}{\sqrt{2}}$

Solution:

$v_1 = \sqrt{2g(h/2)}.$

For v_2 we can replace the liquid of density ρ and height h to a liquid of density 2ρ and height $h/2$.

Thus $v_2 = \sqrt{2gh} = \sqrt{2}v.$



Q.5 Answer:

2.0

Solution:

The linear speed of the liquid at the sides is rw. So, the difference in height is given as follows

$2gh = \omega^2 r^2$

$h = \left[(2 \times 2\pi)^2 \left(5 \times 10^{-2} \right)^2 \right] / (2 \times 10) = 2\ \mathrm{cm}$

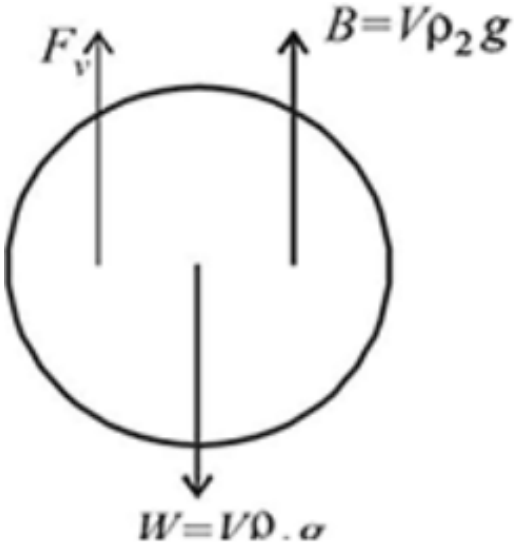
Q.6 Answer:

$\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$

Solution:

@IITJEE_Advanced

The condition for terminal speed (v_t) is Weight = Buoyant force + Viscous force



$\therefore V\rho_1 g = V\rho_2 g + kv_t^2$

$\therefore v_t = \sqrt{\frac{Vg(\rho_1-\rho_2)}{k}}$

Q.7 Answer:

$\frac{8}{9}X$

Solution:

Fluid resistance is given by $\mathrm{R} = \frac{8\eta \mathrm{L}}{\pi \mathrm{r}^4}$

When two capillary tubes of same size are joined in parallel, then equivalent fluid resistance is

$$\mathrm{R_S} = \mathrm{R_1} + \mathrm{R_2} = \frac{8\eta \mathrm{L}}{\pi \mathrm{R}^4} + \frac{8\eta \times 2\ \mathrm{L}}{\pi(2\mathrm{R})^4} = \left(\frac{8\eta \mathrm{L}}{\pi \mathrm{R}^4}\right) \times \frac{9}{8}$$

Rate of flow $= \frac{\mathrm{P}}{\mathrm{R_S}} = \frac{\pi \mathrm{PR}^4}{8\eta \mathrm{L}} \times \frac{8}{9} = \frac{8}{9}\mathrm{X}\left[\text{ as } \mathrm{X} = \frac{\pi \mathrm{PR}^4}{8\eta \mathrm{L}}\right]$

Q.8 Answer:

r^4

Solution:

As indicated in the question as the ball enters the water it has atready attained the terminal velocity. Terminal velocity is reached when the viscous force (acting upwards) balances the weight mg of the bell. Therefore, $6x\eta ru = mg$

where $6x\eta ru$ **is the viscous force on the ball. Here** v **its terminal velocity and** η**, the coefficient of viscosity of water. If** ρ **is the density of the material of the ball,**

Wehere $m = \frac{4}{3}\pi^3 p : 6\pi\eta \cup = \frac{4}{3}\pi^3 pz$

or $u = \frac{2r^2 pg}{9n}$

(1)

Now $u^2 - 0 - 2gh \Rightarrow h = \frac{u^2}{2g}$ **(2)**

From (1) and (2) we find that $h \propto r^4$**.**

@IITJEE_Advanced

Q.9 Answer:

$\frac{2}{81}r^4\left(\frac{\rho-1}{\eta}\right)^2 g$

Solution:

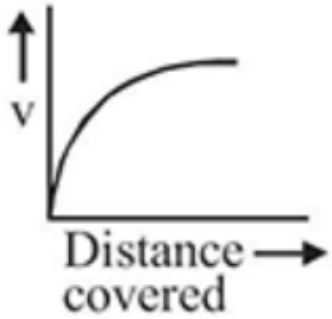
Velocity of ball when it strikes the water surface $v=\sqrt{2gh}$**...(i)**

Terminal velocity of ball inside the water $v=\frac{2}{9}r^2g\frac{(\rho-1)}{\eta}$**...(ii)**

Equation (i) and (ii) we get $\sqrt{2gh}=\frac{2}{9}\frac{r^2g}{\eta}(\rho-1)$

$\Rightarrow h=\frac{2}{81}r^4\left(\frac{\rho-1}{\eta}\right)^2 g$

Q.10 Answer:



Solution:

When a body falls through a viscous liquid, its velocity increases due to gravity but after some time its velocity becomes uniform because of viscous force becoming equal to the gravitational force. Viscous force itself is a variable force which increases as velocity increases, so curve represents the correct alternative.

Q.11 Answer:

$\frac{d_1t_1}{d_2t_2}$

Solution:

According to poiseuille's equation, Volume of liquid flowing per second in a capillary tube is given by

$Q=\frac{V}{t}=\frac{\pi r^4\Delta P}{8\eta L}$

So mass of liquid flowing in a capillary tube is given by

$Q_m=\frac{m}{t}=\frac{\pi r^4\Delta P}{8\eta L}d$

$\Rightarrow \frac{m_1}{t_1}=\frac{\pi r^4\Delta P}{8\eta_1 L}d_1 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(I)$

$\Rightarrow \frac{m_2}{t_2}=\frac{\pi r^4\Delta P}{8\eta_2 L}d_2 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(II)$

Dividing (I) by (II), we have

$\frac{t_2}{t_1}=\frac{d_1}{d_2}\times\frac{\eta_2}{\eta_1} \quad \because m_1=m_2$

$\Rightarrow \frac{\eta_1}{\eta_2}=\frac{d_1t_1}{d_2t_2}$

Q.12 Answer:

0.1

Solution:

@IITJEE_Advanced

Terminal velocity, $v_T = \frac{2r^2(d_1-d_2)g}{9\eta}$

$\frac{v_{T_2}}{0.2} = \frac{(10.5-1.5)}{(19.5-1.5)} \Rightarrow v_{T_2} = 0.2 \times \frac{9}{18}$

$\therefore v_{T_2} = 0.1 \text{ m/s}$

Q.13 Answer:

$\frac{T}{\sqrt{2}}$

Solution:

The velocity of efflux through a hole is given by

$v = \sqrt{2gh}$

where h **is the height of the tank. If the hole is at the top of the tank,** v **is obviously zero. Therefore, the**

average velocity of efflux $v_a = \frac{1}{2} \times \sqrt{2gh} = \sqrt{\frac{gh}{2}}$**. If the volume of water in the tank, when it is full, is** V **and** A **the cross-sectional area of the hole, then time taken by the tank to be emptied is**

$T = \frac{V}{Av_a} = \frac{\sqrt{2}V}{A\sqrt{gh}}$

When the tank is half-full, V **is replaced by** $\frac{V}{2}$ **and** h **is replaced by** $\frac{h}{2}$**. Therefore, time taken now is**

$T' = \frac{\sqrt{2}V/2}{A\sqrt{gh/2}} = \frac{V}{A\sqrt{gh}} = \frac{T}{\sqrt{2}}$

Hence the correct choice is (a).

Q.14 Answer:

$\frac{\sigma}{\rho}$

Solution:

Since, loss of weight = weight of water displaced, the apparent mass of the balloon when completely immersed in water will be m**, which is choice (d).**

Q.15 Answer:

1

Solution:

Refer to Fig.

Velocity of efflux is

$v_1 = \sqrt{2gh_1}$

$\text{and} v_2 = \sqrt{2g\left(H-h_2\right)}$

The times taken by the liquid emerging from A **and** B **to hit the floor are**

$t_1 = \sqrt{\frac{2(H-h_1)}{g}}$

$\text{and} t_2 = \sqrt{\frac{2h_2}{g}}$

Since the two streams strike the floor at the same point $P, R_1 = R_2$ **or** $v_1 t_1 = v_2 t_2$ **or**

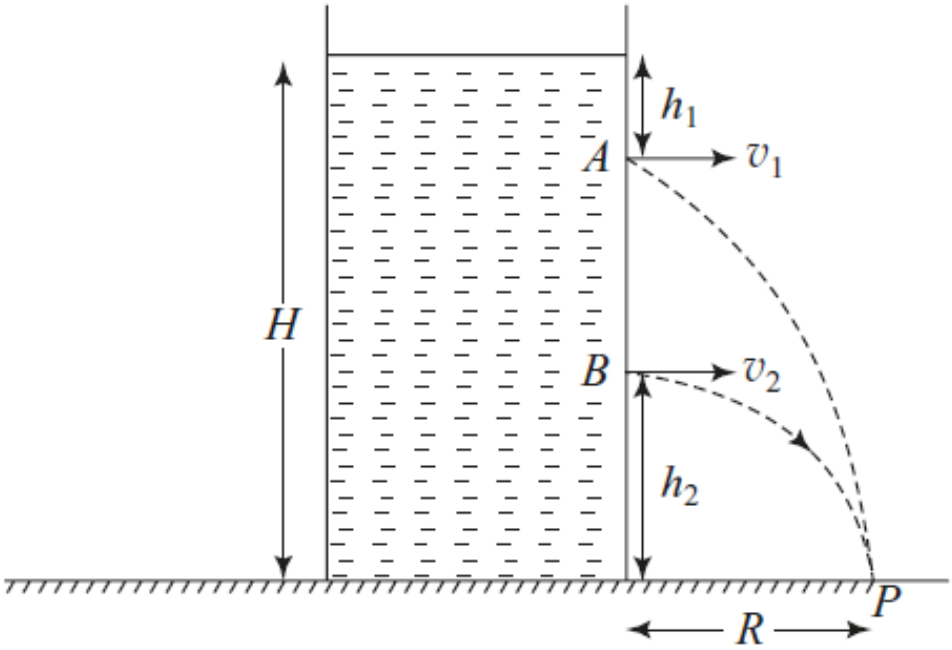
@IITJEE_Advanced

$$\sqrt{2gh_1} \times \sqrt{\frac{2\left(H-h_1\right)}{g}} = \sqrt{2g\left(H-h_2\right)} \times \sqrt{\frac{2h_2}{g}}$$

$$\Rightarrow h_1\left(H-h_1\right) = \left(H-h_2\right)h_2$$

$$\Rightarrow \left[H-\left(h_1+h_2\right)\right]\left[h_1-h_2\right] = 0$$

which gives $H = h_1 + h_2$ (which is not possible) and $h_1 = h_2$. So the correct choice is (d).



@IITJEE_Advanced